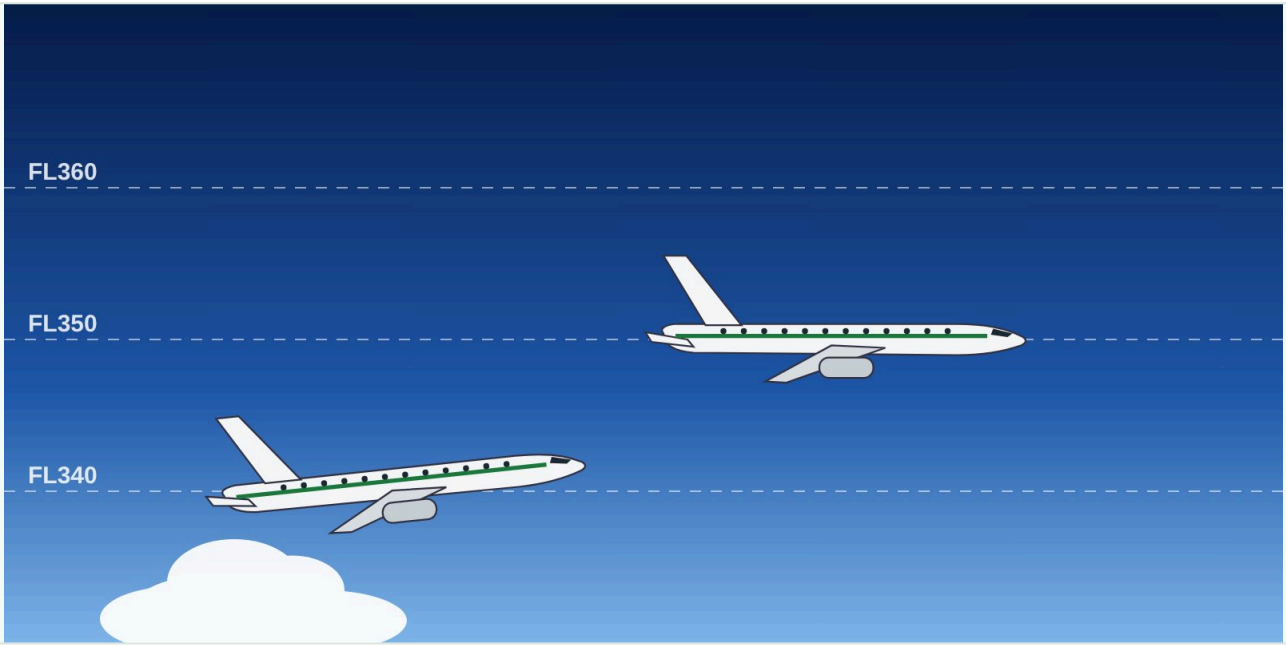
MODULE 2 · CONTINGENCIES, ACAS II AND PHRASEOLOGY

Lesson 05: TCAS / ACAS II in RVSM Airspace

Lesson 5 of 6 · about 6 min

LEARNING OBJECTIVES

- Describe the ACAS II version 7.1 changes.
- Apply procedures that avoid nuisance resolution advisories at adjacent levels.
- Respond correctly to a resolution advisory.



Reduce vertical speed near the cleared level when traffic is at the adjacent level.

ACAS II version 7.1

ACAS II (TCAS II) version 7.1 is the standard required by ICAO and EASA, replacing version 7.0. The main operational change for crews:

- The **"ADJUST VERTICAL SPEED, ADJUST"** resolution advisory has been replaced by **"LEVEL OFF, LEVEL OFF"**, which always requires the vertical rate to be reduced to **0 ft/min**.
- The **reversal logic** has been improved, so that an RA can reverse when the other aircraft does not follow its own RA.

Operate TCAS in **TA/RA** mode during RVSM operations, including in the transition areas.

Avoiding nuisance RAs at adjacent levels

With only 1,000 ft between aircraft, a high vertical rate close to the cleared level can trigger an RA in an aircraft at the adjacent level. ICAO procedures therefore recommend that when a pilot is aware of traffic at an adjacent level (for example from the traffic display), the aircraft climbs or descends at **less than 1,500 ft/min throughout the last 1,000 ft** before the assigned level — especially with the autopilot engaged.

Responding to an RA

- **Follow the RA promptly and accurately**, even if it conflicts with an ATC clearance or instruction. Never manoeuvre in the opposite sense to an RA.
- Tell ATC as soon as possible: "(call sign), TCAS RA."
- When clear: "(call sign), clear of conflict, returning to (assigned clearance)" — then return promptly to the clearance.
- File the company report after any RA.

REMEMBER

A TCAS RA takes precedence over ATC instructions. Following an ATC instruction against an RA has caused mid-air collisions.

KEY TAKEAWAYS

- ACAS II v7.1: "LEVEL OFF, LEVEL OFF" replaces "ADJUST VERTICAL SPEED"; improved reversal logic.
- TA/RA mode in RVSM airspace.
- Less than 1,500 ft/min in the last 1,000 ft when traffic is at an adjacent level.
- Always follow the RA, then inform ATC.

KNOWLEDGE CHECK

What does the ACAS II 7.1 resolution advisory "LEVEL OFF, LEVEL OFF" require?

A. Reduce vertical speed by half

B. Reduce vertical speed to 0 ft/min

C. Climb at 1,500 ft/min

D. Return to the ATC clearance

Answer: B. "LEVEL OFF, LEVEL OFF" always requires the vertical rate to be reduced to 0 ft/min.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.